

Interactive EDA Dashboard

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Abstract—Working with a new dataset usually begins with a set of simple questions: What information is available? How large is the dataset? Are values missing or repeated? Which variables appear related? The Interactive EDA Dashboard developed in this project brings these early checks into one visual interface. Instead of requiring a user to inspect a raw table and create plots one by one, the dashboard presents dataset information, data-quality indicators, and commonly used exploratory graphs in an organized manner. The project supports bar charts, histograms, box plots, scatter plots, and correlation heatmaps. Each view is intended to answer a different exploratory question, such as comparing categories, understanding a numerical distribution, spotting unusually distant observations, or examining relationships between variables. The design is aimed particularly at learners and users who may know the dataset they are working with but may not be comfortable writing analysis code for every initial investigation. The dashboard does not replace analytical judgement; rather, it gives users a convenient starting point from which they can inspect a dataset and decide what deserves further analysis.

Keywords—exploratory data analysis, interactive dashboard, data profiling, data visualization, data quality, histograms, box plots, scatter plots, correlation analysis

I. INTRODUCTION

A raw dataset is rarely easy to interpret at first sight. Even a moderately sized table may contain several variable types, incomplete entries, duplicate rows, and numerical values whose distributions are not obvious from the table itself. Exploratory data analysis (EDA) provides a way to examine these characteristics before moving to statistical modelling or other advanced analysis. Tukey's work established EDA as a practical combination of numerical summaries and graphical investigation [1].

For the present project, the main difficulty is not the absence of visualization software; it is the repeated effort required to perform basic exploratory tasks separately. A user may first inspect the dimensions of a file, then search for null values, check duplicates, choose suitable variables, and finally write separate plotting commands. The Interactive EDA Dashboard combines these steps into a single workflow. The project abstract identifies dataset size and structure, missing and repeated information, and visual exploration as the central requirements [10].

The proposed contribution is therefore a focused EDA interface rather than a new statistical algorithm. Its purpose is to make the first examination of a dataset more

approachable by connecting profiling with five familiar graphical views: bar charts, histograms, box plots, scatter plots, and correlation heatmaps. The same workflow can be reused with different datasets, allowing users to move from a general overview to more specific visual questions without repeatedly rebuilding the analysis from scratch.

II. BACKGROUND AND RELATED WORK

A. Exploratory Data Analysis

Exploratory data analysis is concerned with learning about data before formal modelling. Tukey described EDA as an approach that uses both numerical and graphical methods to reveal features of a dataset [1]. This is useful because a table of summary values may hide patterns that become obvious when the same values are plotted. In the proposed dashboard, this idea is reflected in the combination of dataset summaries and multiple complementary charts.

B. Data Quality and Tidy Structure

Data quality is an important part of early exploration. Wickham's discussion of tidy data emphasizes consistent organization of variables and observations, making subsequent manipulation and visualization easier [2]. In the proposed system, the user is first shown the structure of the supplied dataset and then given indicators for missing and repeated records. These checks are intentionally presented as information for the user rather than automatic corrections, because the appropriate treatment of a missing or repeated value depends on the context of the dataset.

C. Visualization and Dashboards

Python has a mature ecosystem for data analysis and visualization. pandas provides data structures and operations commonly used for preparing and examining tabular data [3], while Matplotlib supplies a general-purpose plotting environment [4]. seaborn extends Python visualization with statistical plotting functions [5]. These tools provide a practical foundation for implementing the charts required by the project. Interactive dashboard frameworks such as Dash can then connect analysis outputs with interface controls, making it possible to explore data through a web-based application rather than through a sequence of notebook cells [6].

A dashboard is useful only when the information can be understood without unnecessary visual effort. Munzner describes visualization as a process of mapping data to visual representations that support analysis [8], while Few emphasizes the importance of presenting dashboard information clearly for rapid interpretation [9]. Following these ideas, the proposed interface places basic dataset information before detailed charts and keeps each graph associated with a recognizable exploratory purpose. The aim is not to display as many graphics as possible, but to provide a small set of views that work together.

III. PROBLEM STATEMENT AND OBJECTIVES

The problem addressed by this project is the difficulty of performing basic EDA efficiently when a user starts with an unfamiliar dataset. Looking through raw rows can make it hard to notice missing information, repeated records, distributions, unusual observations, or relationships between variables. Creating each visualization manually can also be inconvenient for users who are still learning data-analysis techniques. The dashboard addresses this gap by organizing these initial activities into one interactive environment.

A. Objectives

- 1) Give users a quick view of the number of records, variables, and general dataset structure.
- 2) Make missing values and repeated records visible during the initial inspection.
- 3) Provide the five visualization types specified in the project abstract without requiring the user to build every plot manually.
- 4) Help users examine distributions, category comparisons, possible unusual observations, and relationships between numerical variables.
- 5) Keep the workflow simple enough for learners and non-specialist users while still following a standard exploratory-analysis process.

IV. SYSTEM DESIGN

A. Overall Workflow

The system begins with a dataset supplied by the user. After loading, the data is represented in a form that can be inspected programmatically. The dashboard then produces an overview containing dimensions and structural information and performs the basic quality checks required by the project. Once the user has this context, suitable visualizations can be generated from selected variables. The final step is interactive inspection, where the user compares views and decides which observations need additional attention.

The complete workflow can be summarized as: Dataset Upload → Dataset Profiling → Quality Inspection → Chart Selection → Visualization → Interactive Interpretation. This sequence keeps the project focused on exploration. It does not claim that a chart automatically proves a hypothesis; instead, it helps the user find observations that may be worth investigating further.

B. Data Profiling and Quality Checks

The profiling stage gives the user a starting point before any chart is examined. The dashboard should report the number of rows and columns and expose the available variables and their basic structure. These details are useful because they help the user understand what kind of dataset has been supplied and which variables are suitable for different visualizations. For example, categorical variables are more naturally examined through category comparisons, whereas numerical variables can be inspected through distributions and pairwise plots.

Missing values are reported rather than silently discarded. This distinction is important because removing incomplete records without understanding why values are absent can change the dataset. Duplicate records are similarly flagged so that the user can decide whether they represent accidental repetition or legitimate repeated observations. The dashboard therefore treats data-quality checks as part of the exploration stage, leaving the final cleaning decision to the analyst.

C. Visualization Layer

The visualization layer follows the chart types named in the project abstract. Bar charts support comparisons among discrete categories. Histograms divide numerical observations into intervals so that the overall shape of a distribution can be seen [10]. Box plots provide a compact view of central tendency, spread, and observations that lie far from the main range [11]. Scatter plots place two

numerical variables on paired axes and are useful for examining possible trends or clusters. A correlation heatmap summarizes pairwise correlation values in a matrix form, helping users identify variables that move together [12].

These plots are deliberately complementary. A histogram may suggest that a variable is skewed, while a box plot can show whether the same variable contains values far from the central range. A scatter plot can then be used to inspect two variables that appear related, while the heatmap provides a broader view across several numerical variables. The dashboard is intended to help users move between these perspectives rather than treating any single chart as a complete explanation of the data.

D. Interactivity and Usability

Interactivity is used to reduce repetitive work. Depending on the implementation, the user can select variables or chart views and inspect the corresponding visualization without manually rewriting plotting code. Dash provides components for building analytical web applications and connecting interface controls with data-driven outputs [6]. In this project, interactivity is treated as a practical feature: it should make variable selection and visual comparison easier while keeping the interface understandable.

The interface should present information in a logical order. Dataset-level details should be visible before the user begins interpreting individual plots. Chart titles and axis labels should make the selected variables clear, and controls should use familiar terms rather than technical implementation details. This approach is consistent with the general goal of dashboard design: the user should be able to understand what is being displayed and why it is relevant without navigating through unnecessary elements [9].

V. METHODOLOGY

A. Input and Profiling Stage

The methodology starts with a dataset supplied to the application. The system first identifies the dataset dimensions and available fields and then examines basic quality conditions. This stage creates a common information base for the remaining analysis. No conclusion is drawn at this point; the purpose is to establish what the dataset contains and whether any obvious issues need to be considered before visualization.

B. Exploratory Visualization Stage

After profiling, the user selects appropriate visual views. Bar charts are used when the main question involves category-wise comparison. Histograms are selected for studying the distribution of numerical values, while box plots provide another view of spread and potential extreme observations. Scatter plots are used when the analyst wants to inspect two numerical variables together. The correlation heatmap provides a compact summary of pairwise linear association among numerical features. These choices follow established uses of the respective visualization forms [10]–[12].

C. Interpretation Stage

Interpretation remains the responsibility of the user. The dashboard can reveal a possible pattern, but the pattern must be considered in relation to the meaning of the variables and the context in which the data was collected.

For example, a strong correlation does not by itself demonstrate a causal relationship, and an apparent outlier may be a genuine observation rather than an error. The system is therefore designed to support human investigation rather than replace it.

VI. EVALUATION APPROACH AND DISCUSSION

The project abstract does not contain measured experimental results, so this paper does not present invented accuracy values, response times, or user-satisfaction scores. A valid evaluation of the completed dashboard should be based on tests that can be repeated and measured. The evaluation plan below separates functional correctness, visualization correctness, usability, and performance so that each aspect can be assessed independently.

A. Functional Evaluation

Functional testing should begin with valid datasets and verify that the application loads them correctly, reports the number of rows and columns, displays the available variables, and identifies missing and duplicate records when they exist. Additional test files should contain different combinations of numerical and categorical columns so that the visualization controls can be checked under more than one data structure.

B. Visualization Evaluation

Visualization testing should confirm that the selected variables appear in the intended axes or categories and that labels remain readable. Histogram binning should represent the supplied numerical observations, box plots should preserve their statistical summary, scatter plots should use the requested variable pair, and the correlation heatmap should be based on the numerical fields included in the analysis. Testing should also verify that the application responds correctly when an unsuitable variable is selected.

C. Usability Evaluation

Usability can be evaluated with a small group of representative users. Participants can be asked to complete tasks such as finding the dataset dimensions, locating missing values, identifying the distribution of a selected variable, comparing categories, and examining the relationship between two numerical fields. Task completion time, number of errors, and a short user-feedback questionnaire can provide measurable evidence about whether the dashboard is actually easier to use than manual inspection.

D. Discussion

The main strength of the proposed system is its focused workflow. Dataset profiling and visualization are placed together, so a user does not need to repeatedly move between a raw data file and separate plotting commands. This can be particularly useful for students and first-time analysts who are learning what questions to ask during EDA. The project also keeps the visualization set limited, which can make the interface easier to understand than a dashboard containing many unrelated chart types.

There are important boundaries to what the dashboard can conclude. A correlation heatmap describes association and should not be interpreted as evidence of cause and effect. A point outside the main range of a box plot may represent a real event, a measurement issue, or an unusual subgroup. Likewise, a duplicate row is not automatically an error.

These examples show why the dashboard should present evidence clearly while leaving contextual decisions to the analyst.

VII. LIMITATIONS AND FUTURE WORK

The current project scope is centered on introductory and intermediate EDA. The supplied abstract does not specify automated machine-learning models, advanced hypothesis testing, natural-language explanations, distributed processing, or deployment at very large scale. These features should therefore be considered future extensions rather than claimed capabilities of the present system. Another limitation is that no empirical usability or performance results are available yet; those should be collected after implementation.

- 1) Add linked filtering so that a selection in one visualization can update other views.
- 2) Allow users to export selected tables, charts, and summary information for reports.
- 3) Extend the quality-inspection section with additional checks and clearer guidance about possible cleaning actions.
- 4) Add optional statistical summaries while clearly separating exploratory results from formal statistical testing.
- 5) Conduct a user study and report task-completion time, error rate, and user feedback using a defined evaluation procedure.
- 6) Test the application with larger datasets and measure loading time, chart-generation time, and memory usage before considering optimization strategies.

VIII. CONCLUSION

The Interactive EDA Dashboard provides a practical way to begin exploring an unfamiliar dataset. The project brings together dataset profiling, basic quality inspection, and a small set of widely used visualizations in one interface. Its value lies in reducing repetitive setup work and giving users a clear path from a raw dataset to visual inspection. The dashboard is not intended to make analytical decisions automatically; instead, it helps users notice characteristics that deserve further investigation. Future evaluation with real datasets and users will be important for measuring its usability and technical performance. Within its stated scope, the project offers a straightforward and reusable approach to making exploratory data analysis more accessible.

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